

Adnan Harun Dogan

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EDUCATION

-  **Master of Science in Computer Engineering** Sep 2022 – Jan 2025
Middle East Technical University (METU) Ankara, Turkey
 - **Thesis:** [Incorporating Piecewise Linear Functions with Constant Regions in Backpropagation](#)
 - **Advisor:** Prof. Dr. [Sinan Kalkan](#), Assoc. Prof. Dr. [Emre Akbas](#)
-  **Bachelor of Science in Computer Engineering** Sep 2018 – Jul 2022
Middle East Technical University (METU) Ankara, Turkey

EXPERIENCE

-  **Graduate Intern** (Supervisor: Prof. Dr. [Christian Holz](#)) Dec 2024 – Mar 2025
Sensing, Interaction, & Perception Laboratory (SIPLab), ETH Zürich Zürich, Switzerland
 - Developed [learned frequency-weighted Kalman filters](#) that denoise the innovation spectrum under spectral supervision
 - Built [EEG multitaper-spectrum deep models](#) (ConvLSTM) for real-time VR cybersickness detection, +30% over baselines
-  **Research Assistant** (Supervisor: Prof. Dr. [Sinan Kalkan](#), Assoc. Prof. Dr. [Emre Akbas](#)) Feb 2023 – Present
Image Processing and Pattern Recognition Laboratory (ImageLab), METU Ankara, Turkey
 - Contributed to [Bucketed Ranking Loss](#), speeding up ranking-based training of DETR/transformer detectors by up to 6x
 - Developed optimal-transport (Sinkhorn-Knopp) and ranking-based loss formulations for end-to-end object detection
-  **Undergraduate Intern** (Supervisor: Assoc. Prof. Dr. [Erol Sahin](#), Assoc. Prof. Dr. [Bugra Koku](#)) Oct 2021 – Feb 2022
Center for Artificial Intelligence and Robotics (ROMER), METU Ankara, Turkey
 - Built a [self-driving stack](#) for a physical RC car, and trained RL agents for autonomous navigation task in ROS/Gym/PyTorch
 - Simulated a [Hector-quadrotor swarm](#) in ROS/Gazebo, approximating numerical solvers for multi-agent localization
-  **Undergraduate Intern** (Supervisor: Prof. Dr. [Mohammad Abdullah Al-Faruque](#)) Jun 2021 – Sep 2021
Cyber-Physical Systems Laboratory, University of California, Irvine Irvine, California, USA
 - Built [multi-modal deep models](#) for beat-by-beat cardiac and sleep abnormality detection from physiological signals
 - Fused time-series and spectral features for arrhythmia classification, competitive in international challenges

PUBLICATIONS

- Dogan, A. H.**, Demirel, B. U., Holz, C. (2026). *Deep Kalman State Estimation for Drift-Calibrated Full-Body Pose Tracking from Sparse Inertial Sensors*. [\[Under Review\]](#)
- Dogan, A. H.**, Demirel, B. U., Holz, C. (2026). *Magnetometer-Free State Estimation for Sparse Inertial Human Pose Tracking via Motion-Based Yaw Inference*. [\[Under Review\]](#)
- Dogan, A. H.**, Deniz, M., Alemdar, H., Baran, O. U. (2026). *CoFINN: Conservation Flux Informed Neural Networks for Physics Problems Governed by Conservation Laws*. [\[Under Review\]](#)
- Dogan, A. H.**, Demirel, B. U., Holz, C. (2026). *Frequency-Weighted Neural Kalman Filters*. **ICRA 2026**. [\[Code\]](#)
- Demirel, B. U., **Dogan, A. H.**, Rossie, J., Möbus M., Holz, C. (2025). *Beyond subjectivity: Continuous cybersickness detection using eeg-based multitaper spectrum estimation*. **IEEE/TVCG 2025**. [\[DOI | Code\]](#)
- Yavuz, F., Cam B.C., **Dogan, A. H.**, Oksuz, K., Kalkan, S., Akbas, E. (2024). *Bucketed Ranking Loss for Efficient Ranking-based Training of Object Detectors*. **ECCV 2024**. [\[DOI | PDF | Code\]](#)
- Amin M. A., **Dogan, A. H.**, Kuru E. S., Sever Y., Angin P. (Apr 2024). *Misuse Detection and Response for Orchestrated Microservices Based Software*. **AINA 2024**. [\[PDF | Data\]](#)
- Karagoz, P., Cekin, R. F., **Dogan, A. H.**, Oktay, B., Ozturk, A. U., Tonay, S. T., Tunel, B. M. (2024). *Enhancing Underground Built Heritage Analysis with Text Mining: A Case Study on Cappadocia*. Book Chapter 2024. [\[PDF\]](#)
- Sever Y., **Dogan, A. H.** (Aug 2023). *A Kubernetes dataset for misuse detection*. **ITU Journal 2023**. [\[DOI | PDF | Code\]](#)
- Sever, Y., Ekin, G., **Dogan, A. H.**, Alparslan, B., Gurbuz, A. S., Jabrayilov, V., Angin, P. (Sep 2022). *An Empirical Analysis of IDS Approaches in Container Security*. **IEEE/SRMC 2022**. [\[Best Paper Award\] \[DOI\]](#)
- Demirel, B. U., **Dogan, A. H.**, Al-Faruque, M. A. (2021). *Two Might Do: A Beat-by-Beat Classification of Cardiac Abnormalities using Deep Learning and Domain-Specific Features*. **CinC 2021**. [\[DOI | PDF | Code\]](#)
- Buyukbas, E. B., **Dogan, A. H.**, Ozturk, A. U., Karagoz, P. (Sep 2021). *Explainability in Irony Detection*. **DaWaK 2021**. [\[PDF\]](#)
- Dogan, A. H.**, Dogan, A. (Jun 2021). *An Assembled Deep Learning Approach for Flow Field Prediction*. [\[PDF\]](#)

SCORES, HONORS, & AWARDS

- TOEFL iBT:** 109/120 (R29 L29 S25 W26), ETS Oct 2022
- University Entrance Exam:** ranked 1466th of 2.27M candidates, OSYM, Turkey Aug 2017

TECHNICAL SKILLS & RESEARCH INTERESTS

- Languages & Tools:** 🐍 Python/PyTorch, 📌 C++, 🧩 ROS2, 🦀 Rust
- DevOps & Containers:** [GitHub](#), 🐳 Docker, Singularity, Apptainer
- Differentiable Optimization:** Combinatorial Optimization, Optimal Transport, Deep State Space Models
- Deep Learning:** Computer Vision, Self-Supervised Learning, Probabilistic Deep Learning, Deep Reinforcement Learning